

Math 241, F04
Quiz 2

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the position vector $\mathbf{r}(t) = \sin 3t\mathbf{i} - \cos 3t\mathbf{j}$,

a) Find the velocity $\mathbf{v}(t)$, speed $|\mathbf{v}(t)|$ and acceleration $\mathbf{a}(t)$.

Solution:

$$\begin{aligned}\mathbf{v}(t) &= 3 \cos 3t\mathbf{i} + 3 \sin 3t\mathbf{j} \\ |\mathbf{v}(t)| &= \sqrt{(3 \cos 3t)^2 + (3 \sin 3t)^2} = \sqrt{9} = 3 \\ \mathbf{a}(t) &= -9 \sin 3t\mathbf{i} + 9 \cos 3t\mathbf{j}\end{aligned}$$

b) Find $D_t[\mathbf{r}(t) \cdot \mathbf{r}'(t)]$.

Solution:

$$\begin{aligned}\mathbf{r}(t) \cdot \mathbf{r}'(t) &= (\sin 3t\mathbf{i} - \cos 3t\mathbf{j}) \cdot (3 \cos 3t\mathbf{i} + 3 \sin 3t\mathbf{j}) \\ &= \sin 3t(3 \cos 3t) + (-\cos 3t)(3 \sin 3t) \\ &= 0\end{aligned}$$

So we have

$$D_t[\mathbf{r}(t) \cdot \mathbf{r}'(t)] = 0$$

Problem 2. (10 points) If two vectors $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + z\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}$ are perpendicular, find z . (Hint: if $\mathbf{u}$ and $\mathbf{v}$ are perpendicular, then $\mathbf{u} \cdot \mathbf{v} = 0$.)

Solution: $\mathbf{u}$ and $\mathbf{v}$ are perpendicular, so $\mathbf{u} \cdot \mathbf{v} = 0$ which means

$$(2\mathbf{i} + 3\mathbf{j} + z\mathbf{k}) \cdot (2\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}) = 0$$

$$2 \cdot 2 + 3 \cdot 6 - 3z = 0$$

$$3z = 22$$

$$z = \frac{22}{3}$$